FLTK Tutorial
for FLTK v1.3.0 – 1.5.x

This tutorial can be used by the absolute FLTK beginner. In the course of the tutorial the most common widgets will be explained and you will gain a good overview of the FLTK toolkit. The examples have been kept as short and simple as possible so it will not take a lot of time to understand them.

I ported FLTK to DOS and took the screenshots in DOS. The examples will work without any changes on any platform FLTK can be used for.

Contents

1. Introduction

2. The FLTK structure

3. Installation of the FLTK toolkit

4. The first program

5. The button widget

6. Callback functions

7. Input and Output boxes

8. The editor widget

9. The browser widget

10. The menubar widget

11. Toolbar and drop-down list

12. A dialog window with radio buttons

13. Displaying images

14. Grouping widgets in Tabs

15. Handling mouse events part1

16. Handling mouse events part2

17. Displaying the events generated by FLTK

18. The tree widget

19. References

20. Changes

1. Introduction

1.1 FLTK 1.3.0 Tutorial

FLTK shall stand for Fast Light ToolKit and is a GUI written in C++ for C++ programs. It can be used on Linux, Windows, Apple Mac and DOS. FLTK has lower memory requirements than other toolkits and can be linked as a static library with an application. It can also be compiled as a shared library too.

FLTK has been in development over many years and the currently latest version is 1.3.0. There had been a version 2.0 before but unfortunately this branch never got out of alpha and development of that has been canceled now in favor of version 1.3.0. So 1.3.0 is the version any new development should be based on.

6th of March 2012 - Georg Potthast

1.2 FLTK Tutorial, for FLTK 1.3.0 – 1.5.x

The tutorial that Georg created in 2012 has stand the test of time unchanged for more than 14 years, clearly providing information at a level that is asked for and appreciated by many. A remarkable thing to do in a field where monthly updates are not uncommon.

As time passed by FLTK has changed and C++ has addopted new standards. That reflects in the tutorial examples and to address it propperly an update is required. Georg has kindly permitted me to make the changes.

What is changed is the example source code. The text is not changed (unless the source changes forces it). FLTK v1.4.5 is the reference version for the current tutorial, v1.5 is the logical „next step“. All examples are tested against FLTK v1.4.5 and v1.5.x[1] and build without problems. All code is tested for C++11 or better (up to C++23).

All hyperlink referenced are checked and updated (where applicable with FLTK 1.4.5 as reference). A summary on what is changed is available in the appendix „Changes“.

I would like to thank Georg Potthast for allowing me to work with his tutorial. And any mistakes I introduced in changing it are on my account. I also would like to thank the FLTK work group. When asked any question I received advise and invaluable insights into FLTK and its history.

18 sept 2026 - Andre Steenveld

1 FLTK v1.5.x is in development and committed to backwards compatibility. Still changes in it may break tutorial code. No long-term guarantee is given that all tutorial code build against v1.5 untill the development is finished.

1.3 The small prints.

Because it is needed.

Copyright:
Georg Potthast, 6 mar 2012 – 10 sept 2026,
Georg Potthast and others, 11 sept 2026 – …

Maintainer: Andre Steenveld.

FLTK:
reference version: v1.4.5
covering FLTK v1.3.0 – v1.5rev2026/09/10

Version information:
- v1, 6 mar 2012, released as „FLTK 1.3.0 Tutorial“.
- v2, 18 sept 2026, released as „FLTK Tutorial, for FLTK v1.3.0 – 1.5.x“

2. The FLTK structure

`Source paragraphs` and in text `code references` are displayed in a `different font`.

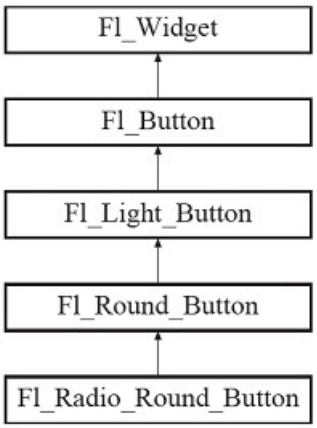
Being written in C++ FLTK defines a class for every widget. GUI toolkits usually refer to all sorts of objects, like buttons, input boxes, scrollbars etc as widgets.

The base class for all widgets is the `Fl_Widget` class. All widgets are subclasses from the `Fl_Widget` class or again subclasses from these subclasses.

This are some of the direct subclasses of `Fl_Widget`: `Fl_Box`, `Fl_Button`, `Fl_Group`, `Fl_Input_`, `Fl_Menu_`, `Fl_Progress` and `Fl_Timer`.

A subclass of `Fl_Group` again is `Fl_Window`, a class you will be using in every program.

An example of a long tree of subclasses is `FL_Radio_Round_Button`. As shown below this button type is a subclass of `Fl_Round_Button` which is a subclass of `Fl_Light_Button` which is again a subclass of `Fl_Button`.



The `FL_Radio_Round_Button` class has access to all methods or functions defined in one of the classes it is a subclass of. You will have to check in the documentation whether any of these classes has a method which does what you want to achieve.

There is also the `FL global` (static) class containing state information and global methods for the current application.

In addition several non-class functions and symbols are provided, they are grouped into header files with lower-case names.

Note: (valid for all examples by default) FLTK provides several options for resizing a window (in the absolute sence). Overall size and appreances can be controled with `<Ctrl>+` and `<Ctrl>-`. There is also an environment variable `FLTK_SCALING_FACTOR` that can be used for a starting value, usualy in the range of 1.0 to 2.0.

3. Installation of the FLTK toolkit

a.) Linux

download the latest source, unpack it and then do configure,make,install. Then compile your program e.g. as:
"fltk-config' --compile example.cxx"
or if you use images:
"fltk-config --use-images --compile example.cxx"

Further details can be found in the FLTK manual here:
https://www.fltk.org/doc-1.4/basics.html There is a section called "Compiling Programs with Standard Compilers" further below in that section.
You can also read the README.Unix.txt file that is included in the FLTK source code package. This also covers using FLTK with the Code::Blocks IDE.

b.) Windows

There are a number of alternatives to develop FLTK programs on Windows. The FLTK site just has the source code so you either compile FLTK on Windows or download binaries from the net as described below.

b1.) DEV-CPP
This alternative is no longer available.

This is an IDE which uses MinGW to generate C and C++ programs on Windows. You can download it here: https://www.dev-cpp.com/Dev-Cpp-5/

~~Then you can download a "devpak" of FLTK 1.3.0 here:~~
~~http://nanox-microwindows-nxlib-fltk-for-dos.googlecode.com/files/fltk-130-1gp.DevPak~~[2]

b2.) Code::Blocks

This is another good IDE using MinGW. If you open a new project, you can select the FLTK icon in there. After that you will have enter the location of the FLTK files. You can download the binaries here: http://code.google.com/p/fltkwinbin/downloads/list
and then put them into a directory of your choice. Then enter that path into the Code::Blocks' input box. Code::Blocks just needs the path to the root FLTK directory where bin, include, lib, share and test are in.

You can then select file->new->empty file and save that as main. Then cut and past one of the examples in this tutorial into the edit window and save the file.

Then right-click on the project, select "Build options" and add the following libraries to the linker settings:
libcomctl32.a, libcomdlg32.a, libole32.a,libgdi32.a and libuuid.a. You find these libraries in the "CodeBlocks\MinGW\lib\" directory.

After this select Build->Run or CTRL-F10 that should build the project and run it.

2 2026/09/10: 404 page not found, no alternative sources.

b3.) Compile FLTK on Windows with MinGW.

~~Both DEV-CPP and~~ Code::Blocks install MinGW on your PC. If you install MinGW after you already have installed one of these IDEs Windows will point to the new MinGW directory ~~and and that causes problems with DEV-CPP~~.

Details how to compile FLTK with MinGW are in the file README.MSWindows.txt which is included in the FLTK source code package.
It also described in there how to use FLTK with Cygwin.

You can also compile FLTK on Windows with Code::Blocks if you follow this tutorial:
http://gintasdx.althirius-studios.com/2011/08/tutorial-codeblocks-with-fltk.html

b4.) Microsoft Visual C++

This is covered in the file README.MSWindows.txt.

Further details can also be found here:
~~http://research.cs.wisc.edu/graphics/Courses/559-f2009/Main/Tutorial1~~ [3]

c.) Apple OS X

Please read the file README.OSX.txt which is included in the FLTK source code package.

d.) DOS

Download the DJGPP[4] package from
http://code.google.com/p/nanox-microwindows-nxlib-fltk-for-dos/downloads/list ,
edit the start.bat file in there to your paths. The required FLTK libraries are already included in the DJGPP package but can be downloaded from that site separately as well.

You can compile the examples either with this line (observe the line wrap here):

```
gpp -g -I/djgpp/include -o example.exe example.cxx -L/djgpp/lib -lfltk -lNX11 -lnano-X
-lfreetype
```

For the examples which load an image you have to add additional parameters to this line:

```
gpp -g -I/djgpp/include -o ex13.exe ex13.cxx -L/djgpp/lib -lfltk_images -lfltk -lNX11 -lnano-X
-lfreetype -ljpeg -lpng -lz
```

Or you enter a shell with "sh" and use the "fltk-config" script:

```
sh-2.04$ 'fltk-config' --compile example.cxx
sh-2.04$ 'fltk-config' --use-images --compile example.cxx
```

3 2026/09/10: site discontinued.
4 2026/09/10: package still available, usability not tested.

4. The first program

This tutorial starts with very simple examples. Some readers may consider them too simple but those may read these parts more quickly.



Our first program just opens a window with a title. You can terminate the program by closing the window with your mouse or pressing the ESC key.

To run this or any other example please copy the lines below and paste them into the editor that you will use for compiling the code.

```
example1.cpp: """
#include <FL/Fl.H>
#include <FL/Fl_Window.H>
constexpr const int WIDTH = 400;
constexpr const int HEIGHT = 400;
int main() {
  // Create a window - width, height, label (=title)
  Fl_Window win(HEIGHT, WIDTH,"FLTK Tutorial - Example 1");
  // Display the window
  win.show();
  // Run and return
  return Fl::run();
}
"""
```

All programs must include the file `<FL/FL.H>` to include the FLTK global class FL. In addition the program must include a header file for each FLTK class it uses, in our case it is just `Fl_Window`, a subclass of `Fl_Group`.

The statement:

```
  Fl_Window win(HEIGHT, WIDTH,"FLTK Tutorial - Example 1");
```

creates a new object of the `Fl_Window` class called "win". The call also defines the size of the window to be created and the title string.

Then there is a call the "show" method of the `Fl_Window` class to display the window:

```
  win.show();
```

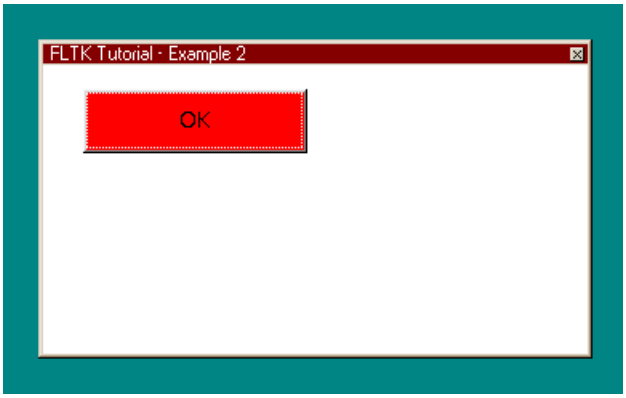
Finally every FLTK program has to have the `Fl::run()` statement:

```
  return Fl::run();
```

This enters the FLTK event loop. The program now waits for events, like mouse clicks or keystrokes to happen and act upon them. When the window is closed or the ESC key pressed, this function will return.

5. The button widget

Here are two examples that add a button to the window made in the example above. The first one extends the example above while the other defines a subclass of `FL_Window` which gives the code a completely different structure.




```
example2.cpp: """
#include <FL/Fl.H>
#include <FL/Fl_Window.H>
#include <FL/Fl_Button.H>
constexpr const int WIDTH = 340;
constexpr const int HEIGHT = 180;
int main(int argc, char **argv) {
  // Create a window - width, height, label (=title)
  Fl_Window *win = new Fl_Window(WIDTH,HEIGHT,"FLTK Tutorial - Example 2");
  // Set color of window to white
  win->color(FL_WHITE);
  // Begin adding children to this window
  win->begin(); // recomended habit: intent between begin() and end()
    // Create a button - x , y , width, height, label.
    // Implicit: win takes ownership of button1.
    Fl_Button *button1 = new Fl_Button(25,15,140,40,"OK");
    // Set color of button to red
    button1->color(FL_RED);
    // Stop adding children to this window
  win->end();
  // Display the window
  win->show();
  // Run and return
  auto result = Fl::run();
  // clean up.
  delete win; // implicit: also delete button1.
  return result;
}
"""
```

Since we are using a new widget here, the button widget, we have to include a header file for that class first.

The statement:

```
  Fl_Window *win = new Fl_Window(WIDTH,HEIGHT,"FLTK Tutorial - Example 2");
```

creates a pointer to the new window object "win". This is different to the first example and we will have to use "`win->show();`" now instead of "`win.show();`" as before.

Then we use a "set" method to turn the color of our window to white:

```
  win->color(FL_WHITE);
```

FLTK usually has a corresponding "get" method for each "set" method. Here "`Fl_color c = win->color();`" would return the current window color in "c".

Then there is:

```
  win->begin();
```

that tells FLTK to start a group of children for our "win" object. This statement could be omitted and FLTK will add this implicitly then.

Following that we create a pointer to a new button object:

```
  Fl_Button *button1 = new Fl_Button(25,15,140,40,"OK");
```

and make a "set" statement to set its color to red. Instance win will implicitly take control of the button. On destruction of win it will also delete the button1 instance. Using new and not using delete in the same scope may feel awkward at first. It is the way FLTK handles it.

The statement

```
  win->end();
```

will tell FLTK that we are done defining the children for the "win" object. The defined group will be set to the parent of the window, in this case to `NULL` because "win" does not have a parent.

The following statements will cause the window to be displayed and enter the FLTK event loop. Before returning proper cleanup is done for win. That instanse also owns button1 and will delete it too.

So far we have not defined classes in the examples. To show how this can be done using FLTK classes and inheritance we rewrite the example above:

example3.cpp: """
```
#include <FL/Fl.H>
```

```
#include <FL/Fl_Window.H>
#include <FL/Fl_Button.H>
constexpr const int WIDTH = 340;
constexpr const int HEIGHT = 180;
class MyWindow : public Fl_Window {
public:
  MyWindow(int width, int height, const char* title=0) :
    Fl_Window(width,height,title) {
    // Set color of window to white
    color(FL_WHITE);
    // Begin adding children to this window
    begin();
      // Create a button - x , y , width, height, label
      // Implicit: this instance parent takes ownership of button1.
      Fl_Button *button1 = new Fl_Button(25,15,140,40,"OK");
      // Set color of button to red
      button1->color(FL_RED);
      // Stop adding children to this window
    end();
    // Display the window
    show();
  }
};
int main() {
  // Create a window with our new class - width, height, label (=title)
  MyWindow win(WIDTH,HEIGHT,"FLTK Tutorial - Example 2++");
  // Run and return, implicit cleanup of win.
  return Fl::run();
}
"""
```

Here the statement

```
  class MyWindow : public Fl_Window
```

creates a class called `MyWindow` which is derived from the `Fl_Window` class. In the constructor of this class we define the color of the window, the button as a child of the window and display the window. Since the "`this`" pointer is implicit, you do not need a pointer in front of the `begin()`, `end()` and `show()` statements.

In the main function we just create a new object of the `MyWindow` class called "win" and then enter the FLTK event loop.

6. Callback functions

Now we will add a callback function which will be called by FLTK as soon as the button is clicked. This callback function will then change the color of the button.

There is no screenshot provided since it looks like the one in the previous example.

```
example4.cpp: """
#include <FL/Fl.H>
#include <FL/Fl_Window.H>
#include <FL/Fl_Button.H>
constexpr const int WIDTH = 340;
constexpr const int HEIGHT = 180;
void button1_cb(Fl_Widget* buttonptr) {
  if (buttonptr->color() == FL_BLUE) {
    buttonptr->color(FL_RED); //toggle
  } else {
    buttonptr->color(FL_BLUE); //toggle
  }
}
int main(int argc, char **argv) {
  // Create a window - width, height, label (=title)
  Fl_Window *win = new Fl_Window(WIDTH,HEIGHT,"FLTK Tutorial - Example 4");
  win->color(FL_WHITE);
  // Begin adding children to this window
  win->begin();
    //Create a button - x , y , width, height, label
    Fl_Button *button1 = new Fl_Button(25,15,140,40,"Click me!");
    button1->color(FL_RED);
    //register callback function with this button
    button1->callback(button1_cb);
    // Stop adding children to this window
  win->end();
  // Display the window
  win->show();
  // Run
  auto result = Fl::run();
  // clean up.
  delete win; // implicit: also delete button1.
  // and return
  return result;
}
"""
```

The statement:

```
  button1->callback(button1_cb);
```

defines that the callback "`button1_cb`" will be called when the user clicks on it.

This function is defined as

```
  void button1_cb(Fl_Widget* buttonptr)
```

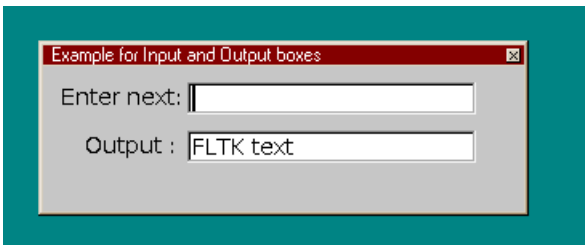
Please observe that a pointer to the `Fl_Widget` class is passed and not to `Fl_Button`. This way we can only use methods defined in `Fl_Widget`. For this example this is fine since we will only be using the "color" method defined in `Fl_Widget`. If we want to use methods defined in the `Fl_Button` class we will have to cast the buttonptr to a `Fl_Button` class e.g:

```
  Fl_Button* b = (Fl_Button*)buttonptr;
```

The rest of the function checks if the button color is blue and if yes turns it to red and vice versa.

7. Input and Output boxes

This example will display an input and an output widget. The text you enter in the input box will be copied into the output box and the label of the input box will be changed.




```
example5.cpp: """
#include <FL/Fl.H>
#include <FL/Fl_Window.H>
#include <FL/Fl_Input.H>
#include <FL/Fl_Output.H>
constexpr const int WIDTH = 300;
constexpr const int HEIGHT = 90;
// Fl_Input input1(90, 10, 180, 20, "Input : "); // will fail
// Fl_Output output1(90, 40, 180, 20, "Output : "); // will fail
Fl_Input *input1 = nullptr;
Fl_Output * output1 = nullptr;
static void cb_input1(Fl_Input*, void* userdata) {
  input1->label((const char*)userdata);
  output1->value(input1->value());
  input1->value(""); //clear again
}
int main(int argc, char **argv) {
  Fl_Window win(WIDTH,HEIGHT,"Example for Input and Output boxes");
  win.begin();
    // win will take ownership of *input1;
    //win.add(input1); will fail during cleanup of win.
    input1 = new Fl_Input (90, 10, 180, 20, "Input : ");
    input1->callback((Fl_Callback*)cb_input1,(void *)"Enter next:");
    input1->when(FL_WHEN_RELEASE | FL_WHEN_ENTER_KEY);
    // win will take ownership of *output1;
    //win.add(output1); will fail during cleanup of win.
    output1 = new Fl_Output(90, 40, 180, 20, "Output : ");
  win.end();
  win.show();
  // cleanup of win: delete *input1 which fails with an application termination
```

```
  // if it points to a static instance. (can't delete a static instance)
  return Fl::run();
}
"""
```

Here if `input1` and `output1` are defined as globals they can be accessed in the `input1` callback function. Then there will be a problem with cleanup of win. A delete to static instances usualy result in early program termination. To avoid problems do not use static instances, use global pointers. There is more than one way to avoid this problem.

In the main function we define a window object called "win" and add the `input1` and `output1` widgets to that, not by using the "`win.add()`" statements but by creating a new instance pointed to by the global pointer. And `win` will take ownership of the pointer. There are no problems during cleanup of `win`.

In the next line:

```
  input1->callback((Fl_Callback*)cb_input1,(void *)"Enter next:");
```

the callback function "`cb_input1`" is set to be called if an event regarding the `input1` widget occurs.

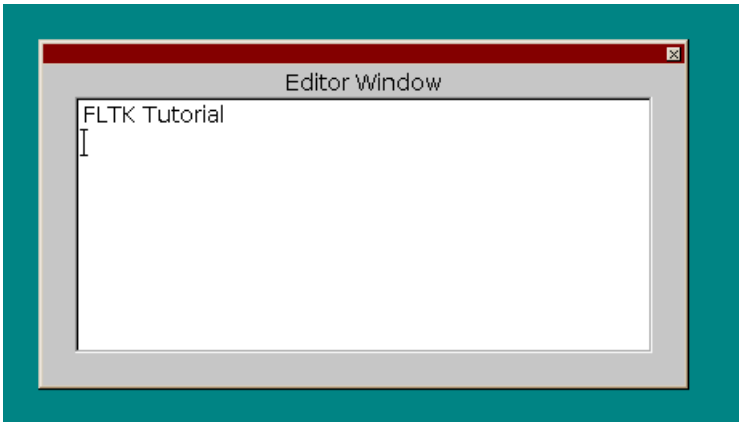
As an example we have used the second parameter of the callback function here to pass a void pointer to the string "Enter next:". This parameter is called the "userdata" parameter. Here the pointer will be used as label text in the callback function.

Following that using the "`.when()`" method it is defined what event shall trigger the callback function. This will be the Enter key and when the input widget is released.

In the callback function the label of the `input1` widget is changed to the userdata string. The value of the input widget, i.e. the text entered into this widget, is copied into the output widget und the input widget is cleared again after that.

8. The editor widget

This program shows how to use the editor widget included in FLTK. This example is taken from Greg Ercolanos FLTK cheat sheet page at: http://seriss.com/people/erco/fltk/



```
example6.cpp: """
#include <FL/Fl.H>
#include <FL/Fl_Window.H>
#include <FL/Fl_Text_Editor.H>
constexpr const int WIDTH = 400;
constexpr const int HEIGHT = 200;
int main() {
  Fl_Window *win = new Fl_Window(WIDTH, HEIGHT);
  Fl_Text_Buffer *buff = new Fl_Text_Buffer();
  Fl_Text_Editor *disp = new Fl_Text_Editor(20, 20, 360, 160, "Editor Window");
  disp->buffer(buff);
  win->resizable(*disp);
  win->show();
  buff->text("FLTK Tutorial"); //add initial text here if required
  auto result = Fl::run();
  return result;
}
"""
```

Although the editor widget is comprised of a lot of FLTK code, it is simple to use. Memory houskeeping needs some attention.

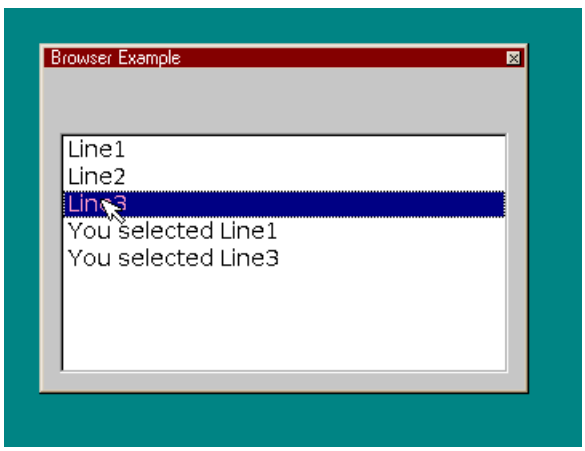
First there are pointers created to the new window object "win", the new buffer object "buff" and the new editor widget "disp". Then the buffer object is attached to the editor object using the "buffer" method the `Fl_Text_Editor` widget has inherited from the `FL_Display` widget:

```
  disp->buffer(buff);
```

With the "`text()`" method we can move a string into the buffer.

9. The browser widget

The browser widget displays an array of strings line by line in a window and lets the user select items from this list. Here the browser is initialized with three lines. Whenever one of these lines is clicked an additional line is added stating what line was selected.



```
example7.cpp: """
#include <FL/Fl.H>
#include <FL/Fl_Window.H>
#include <FL/Fl_Browser.H>
constexpr const int W = 300;
constexpr const int H = 200;
constexpr const int M = 10;
void browser_cb(Fl_Widget *w) {
  Fl_Browser *b = (Fl_Browser*)w; //cast to get access to Browser methods
  // retrieve selected item from browser
  int index = b->value();
  // add text to browser using the retrieved index number
  if ( index == 1 ) {
    b->add("You selected Line1");
  } else if ( index == 2 ) {
    b->add("You selected Line2");
  } else if ( index == 3 ) {
    b->add("You selected Line3");
  }
}
int main() {
  Fl_Window *win  = new Fl_Window(W,H,"Browser Example");
  Fl_Browser *b = new Fl_Browser(M,4*M, win->w()-2*M, win->h()-5*M);
  b->type(FL_MULTI_BROWSER);
  b->add("Line1");
  b->add("Line2");
  b->add("Line3");
  b->callback(browser_cb);
  win->show();
  auto result = Fl::run();
  delete b;
  delete win;
  return result;
}
"""
```

In the `main()` function we first create pointers to a new window and browser object. For the browser object we use the "get" functions "`win->w()`" and "`win->h()`" to retrieve the size of the window "win" to fit it into that. Then we add three lines into the browser object and register its callback function as "browser_cb".

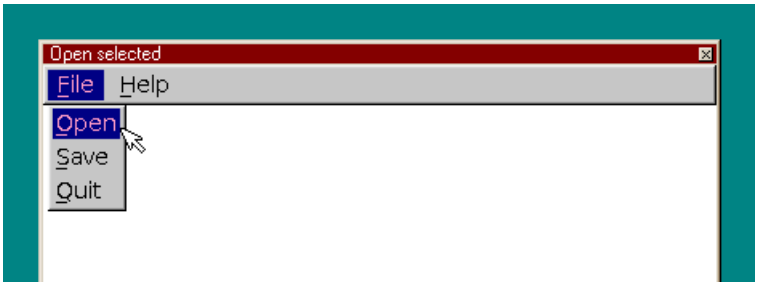
In the callback function we first cast the `Fl_Widget` pointer to a `Fl_Browser` pointer to be able to use the value method of this widget. With the statement:

```
  int index = b->value();
```

we can retrieve which of the three lines the user has clicked last and thus selected it. The index of this line is used to add additional lines to the browser specifying which line had been selected on each click.

10. The menubar widget

This example opens a window with a small menubar. The menu items selected will be displayed in the window title.



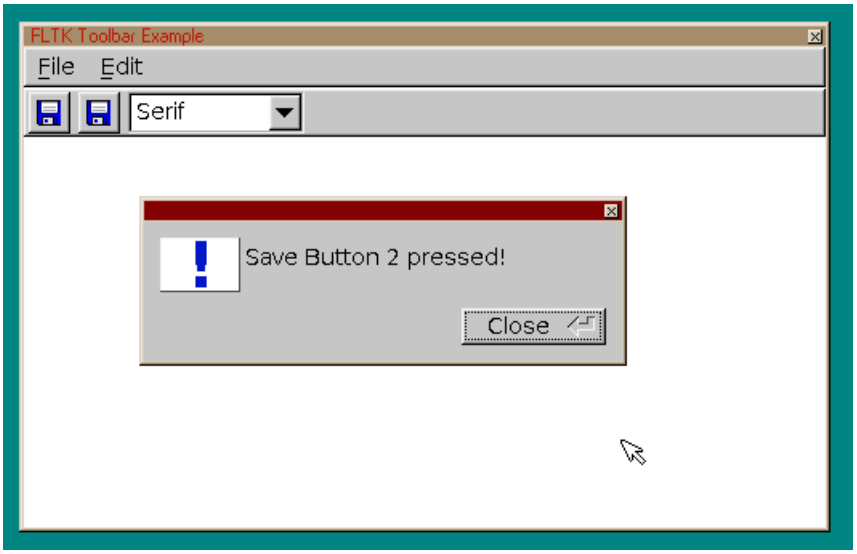

```
example8.cpp: """
#include <FL/Fl.H>
#include <FL/Fl_Window.H>
#include <FL/Fl_Menu_Bar.H>
constexpr const int WIDTH = 420;
constexpr const int HEIGHT = 280;
Fl_Window* win = nullptr;
void Menu_CB_Open(Fl_Widget* w,void*) {win->label("Open selected");}
void Menu_CB_Save(Fl_Widget* w,void*) {win->label("Save selected");}
void Menu_CB_Quit(Fl_Widget* w,void*) {win->label("Quit selected");}
void Menu_CB_Help(Fl_Widget* w,void*) {win->label("Help selected");}
int main() {
  // Open the application window and menu bar with callbacks
  win = new Fl_Window(WIDTH, HEIGHT);
  win->color(FL_WHITE);
  // implicit: win->begin();
    Fl_Menu_Bar * menubar = new Fl_Menu_Bar(0, 0, win->w(), 25);
    menubar->add("&File/&Open", 0, Menu_CB_Open);
    menubar->add("&File/&Save", 0, Menu_CB_Save);
    menubar->add("&File/&Quit", 0, Menu_CB_Quit);
    menubar->add("&Help", 0, Menu_CB_Help);
  win->end();
  win->show();
  auto result = Fl::run();
  delete win;
  win = nullptr;
  return result;
}
"""
```

To keep the example short, "win" is defined as a global `Fl_Window` object pointer so we can use it directly in the callback functions.

Here a `Fl_Menu_Bar` object called "menubar" is added to our main window. There are four menubar items defined. The "&" character defines the shortcut key for this menu item. The first three statements define the "File" menu with the subitems "Open", "Save" and "Quit". The "Help" item has no submenu items. Each statement defines the callback function that shall be called if the user clicks on the menu item. Here these callback functions change the label (title) of the main window to show which menu item had been selected.

11. Toolbar and drop-down list

Many applications place a toolbar with several icons below a menu bar. In this example not only icons but also a drop down list is placed in such a toolbar.




```
example9.cpp: """
#include <FL/Fl.H>
#include <FL/Fl_Window.H>
#include <FL/Fl_Button.H>
#include <FL/Fl_Pixmap.H>
#include <FL/Fl_Input.H>
#include <FL/Fl_Group.H>
#include <FL/Fl_Choice.H>
#include <FL/fl_ask.H>
#include <FL/Fl_Menu_Bar.H>
#include <cstdint>
constexpr const int WIDTH = 420;
constexpr const int HEIGHT = 280;
constexpr const int FONT_SERIF = 1;
constexpr const int FONT_SANS_SERIF = 2;
constexpr const int FONT_MONOSPACE = 3;
static const char *save_xpm[] = {
  "16 16 9 1",
  "      c None",
  ".     c #000000",
  "+     c #0000FF",
  "@     c #000080",
  "#     c #A0A0A0",
  "$     c #0000C0",
  "%     c #FFFFFF",
  "&     c #C0C0FF",
  "*     c #DCDCDC",
  "................",
  ".+@##########@+.",
  ".$@%%%%%%%%%%@$.",
  ".+@%%%%%%%%%%@+.",
  ".+@%#########%@+.",
  ".+@%%%%%%%%%%@+.",
  ".+@%#########%@+.",
  ".+@&%%%%%%%%&@+.",
  ".++@@@@@@@@@@++.",
```

```
  ".+++++++++++++++.",
  ".++$......@@$++.",
  ".++.******.+@++.",
  ".++.*@+***.+@++.",
  ".++.*@+***.+@++.",
  ".@+.******.+@++.",
  " ..............."};
class Toolbar : public Fl_Group {
public:
  Fl_Button *save1;
  Fl_Button *save2;
  Fl_Pixmap *p_save;
  Fl_Choice *font_group;
  Fl_Menu_Item *font_group_items;
  static void cb_save(Fl_Widget*, void*);
  static void cb_fonts(Fl_Widget*, void*);
  Toolbar(int Xpos, int Ypos, int Width, int Height);
};
void Toolbar::cb_save(Fl_Widget *w, void *data) {
  fl_alert(
    "Save Button %d pressed!",
    static_cast<int>(reinterpret_cast<intptr_t>((data)))
  );
}
void Toolbar::cb_fonts(Fl_Widget *w, void *data) {
  fl_alert(
    "Font number %d selected!",
    static_cast<int>(reinterpret_cast<intptr_t>((data)))
  );
}
Toolbar::Toolbar(int Xpos, int Ypos, int Width, int Height) :
  Fl_Group(Xpos, Ypos, Width, Height) {
  box(FL_UP_BOX);
  Ypos += 2; Height -= 4; Xpos += 3; Width = Height;
  int i;
  save1 = new Fl_Button(Xpos, Ypos, Width, Height); Xpos += Width + 5;
  save2 = new Fl_Button(Xpos, Ypos, Width, Height); Xpos += Width + 5;
  font_group = new Fl_Choice(Xpos, Ypos, 110, Height); Xpos += 111;
  p_save = new Fl_Pixmap(save_xpm);
  save1->image(p_save);
  save2->image(p_save);
  save1->tooltip("Save file1");
  save1->callback(cb_save,(void*)1);
  save2->tooltip("Save file2");
  save2->callback(cb_save,(void*)2);
  font_group_items = new Fl_Menu_Item[4];
  for (i = 0; i < 4; i++) font_group_items[i].text = NULL;
  font_group_items->add("Serif", 0, cb_fonts, (void*) FONT_SERIF, 0);
  font_group_items->add("Sans-Serif", 0, cb_fonts,(void*) FONT_SANS_SERIF, 0);
  font_group_items->add("Monospace", 0, cb_fonts, (void*) FONT_MONOSPACE, 0);
  font_group->menu(font_group_items);
}
int main() {
  Fl_Window win(WIDTH, HEIGHT,"FLTK Toolbar Example");
  win.color(FL_WHITE);
  win.begin();
    Fl_Menu_Bar menubar(0, 0, win.w(), 25);
    menubar.add("&File", 0, 0);
    menubar.add("&Edit", 0, 0);
```

```
    Toolbar tool(0,26,win.w(),30);
    tool.clear_visible_focus(); //just use mouse, no TABs
  win.end();
  win.show();
  return Fl::run();
}
"""
```

In the main window first there is defined a menubar and below that an object of the Toolbar class which is defined in this program. When this Toolbar object is created its position and size is specified.

The constructor will first make a box of the size of the toolbar. Then two icons and a drop-down list widget are placed in this box. The icons are defined as an XPM image which again is specified at the top of the code. ~~An XPM image can e.g. be created on Windows with the xpmedit program: http://www.jland.org/swat/xpmedit/~~[5]. To save space the same icon is used here twice in the example.

Then an `FL-Pixmap` object of the XPM image is made in the constructor and using `image()` it is set as the image for both buttons. Further for each button a tooltip and a callback function is specified.

Following that the drop-down list is defined. The items of this list are specified as menu items. These items have the same callback function, just different userdata is passed. After calling `menu()` the items are shown in the drop-down list on the screen.

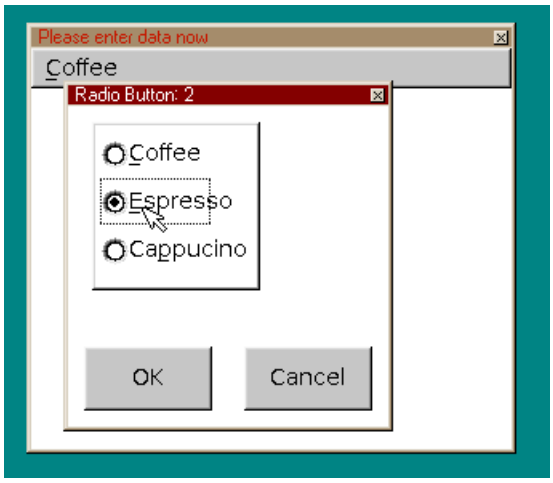
The callback functions use the "`fl_alert`" function to display the userdata passed and indicate which button or item has been selected by the user. The "`fl_alert`" function can take the message text in an sprintf-like format.

FLTK features a number of common dialog functions which are documented in the "modules" section of the FLTK documentation. These are defined with lower-case letter names.

5 2026/09/10: 404, not found. Alternative: Use GIMP or any other graphical tool capable for the XPM file format.

12. A dialog window with radio buttons

This example shows how to open a dialog window when a menu item is selected. It is often necessary to define a window where the user can make a number of selections.




```
example10.cpp: """
#include <FL/Fl.H>
#include <FL/Fl_Group.H>
#include <FL/Fl_Window.H>
#include <FL/Fl_Button.H>
#include <FL/Fl_Round_Button.H>
#include <FL/Fl_Menu_Bar.H>
#include <cstdint>
constexpr const int WIDTH = 300;
constexpr const int HEIGHT = 250;
class Dialog_Window : public Fl_Window {
public:
  int status;
  int radio;
  static Dialog_Window *dw_this;
  static void cb_button(Fl_Widget *w, void *d);
  static void cb_radio(Fl_Widget *b, void *d);
  Dialog_Window(
    int xpos, int ypos, int width, int height,
    const char* title=0
  ) : Fl_Window(xpos,ypos,width,height,title) {
    dw_this = this;
    color(FL_WHITE);
    begin(); // Begin adding children to this window
      Fl_Group* rb_group = new Fl_Group(15, 10, 105, 105);
      // implicit: rg_group->begin();
        rb_group->box(FL_UP_FRAME);
        {
          Fl_Round_Button* rb1 = new Fl_Round_Button(20, 15, 70, 30, "&Coffee");
          rb1->type(102);
          rb1->down_box(FL_ROUND_DOWN_BOX);
          rb1->callback(cb_radio,(void*)1);
        }
        {
          Fl_Round_Button* rb2 = new Fl_Round_Button(
            20, 45, 70, 30, "&Espresso"
```

```
          );
          rb2->type(102);
          rb2->down_box(FL_ROUND_DOWN_BOX);
          rb2->callback(cb_radio,(void*)2);
        }
        {
          Fl_Round_Button* rb3 = new Fl_Round_Button(
            20, 75, 70, 30,"Ca&ppucino"
          );
          rb3->type(102);
          rb3->down_box(FL_ROUND_DOWN_BOX);
          rb3->callback(cb_radio,(void*)3);
        }
      rb_group->end();
      Fl_Button *button_ok = new Fl_Button(10,150,80,40,"OK");
      Fl_Button *button_c = new Fl_Button(110,150,80,40,"Cancel");
      button_ok->callback(cb_button,(void*)1);
      button_c->callback(cb_button,(void*)2);
    end();
    set_modal();
    show();
  }
};
Dialog_Window *Dialog_Window::dw_this = NULL;
void Dialog_Window::cb_radio(Fl_Widget *b, void *d) {
  char msg[64];
  dw_this->radio=static_cast<int>(reinterpret_cast<intptr_t>((d)));
  sprintf(msg, "Radio Button: %d", dw_this->radio);
  dw_this->label(msg);
}
void Dialog_Window::cb_button(Fl_Widget *w, void *d) {
  dw_this->hide();
  dw_this->status = static_cast<int>(reinterpret_cast<intptr_t>((d)));
}
void cb_file(Fl_Widget *w, void *data) {
  Fl_Window* parent=static_cast<Fl_Window*>(data);
  parent->label("Please enter data now");
  // side effect: dw .ctor calls show() and is modal. (compiler may warn for unused var)
  Dialog_Window* dw = new Dialog_Window(
    parent->x()+20, parent->y()+20, 200, 200, NULL
  );
}
int main() {
  Fl_Window win(WIDTH, HEIGHT,"FLTK Dialog Window Example");
  win.color(FL_WHITE);
  win.begin();
    Fl_Menu_Bar menubar(0, 0, win.w(), 25);
  menubar.add("&Coffee", 0, cb_file,(void*)&win);
  win.end();
  win.show();
  return Fl::run();
}
"""
```

In this example the main window just has a menu bar with one item to call the dialog window. The menu item passes a pointer to the main window as userdata in the callback. The callback can then change the label of the main window and read its position so the dialog window is created at a fixed position relative to the main window no matter where

this currently is on the screen. The callback makes an object of the `Dialog_window` class called "`dw`". This class defines the dialog window called `Dialog_window`.

The constructor of this class opens the dialog window and defines a group with three radio buttons or round buttons. There is a box drawn around the group of radio buttons.
Also there are an OK and a Cancel button defined in the dialog window. The dialog window is opened "modal" so user input is directed to this window only while it is displayed.

The callback functions of each button are passed the number of the button in the userdata parameter. Thes callback function makes a msg string with this number and writes it to the title of the dialog window. It also writes the number to the public integer variable "radio" in the `Dialog_window` object.

The OK and Cancel button callbacks hide the dialog window and write a status value into the public variable of the `Dialog_window` object.

This example also shows how to make the "`this`" pointer of the `Dialog_window dw` object available in the entire program. The "`this`" pointer of this object is written into the static variable "`dw_this`" which is also initialized outside the class definition and therefore can be accessed from anywhere in the program.

13. Displaying images

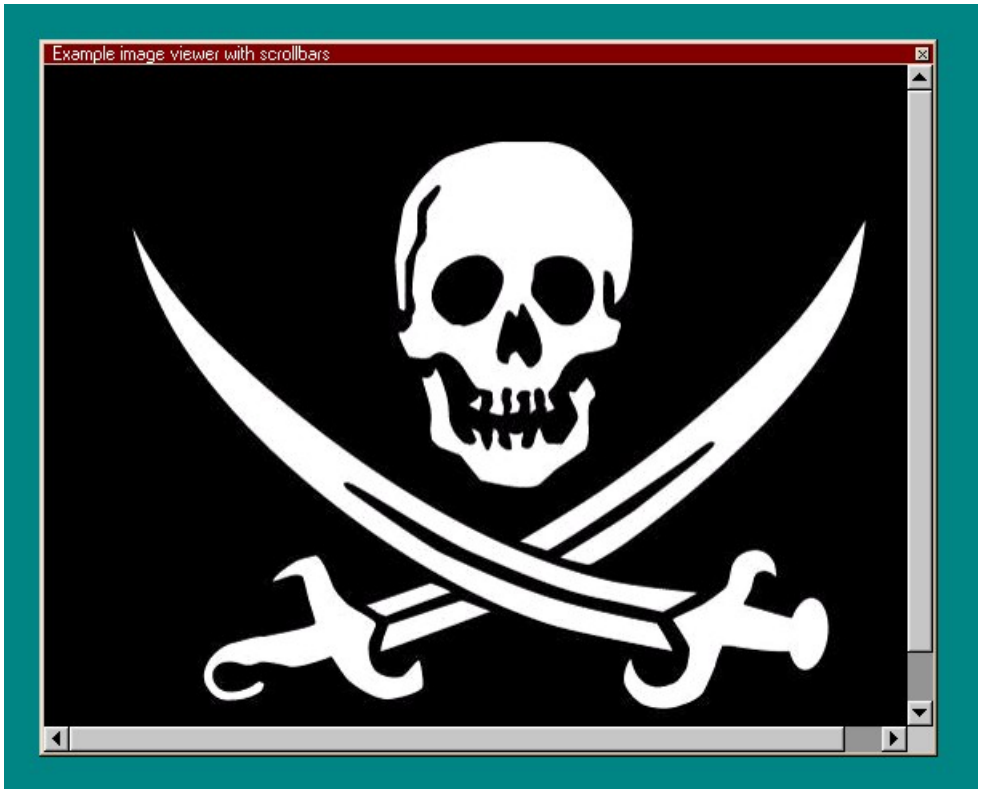
The first example displays a PNG image in a window, the second displays a JPG image in a smaller window and lets you scroll the image using scrollbars. Both examples were developed by Greg Ercolano.

```
example11.cpp: """
#include <FL/Fl.H>
#include <FL/Fl_Window.H>
#include <FL/Fl_Shared_Image.H>
#include <FL/Fl_PNG_Image.H>
#include <FL/Fl_Box.H>
constexpr const int WIDTH = 500;
constexpr const int HEIGHT = 400;
constexpr const int MARGIN = 10;
constexpr const char * PNGFILE = "jolly.png"; // any .png (500 x 400)px will do.
int main() {
  fl_register_images();          // initialize image lib
  Fl_Window win(WIDTH,HEIGHT);   // make a window
  Fl_Box box(MARGIN,MARGIN,WIDTH-2*MARGIN,HEIGHT-2*MARGIN);
                                 // widget that will contain the image
  Fl_PNG_Image png(PNGFILE);     // load png image into ram
  box.image(png);                // attach png image to box
  win.show();
  return(Fl::run());
}
"""
```

The included comments explain the example already. There is a new widget used here: `Fl_Box`. This box is not visible since when you attach the image to this object you see the

image in the defined box.

When creating the `Fl_PNG_Image` object the image file is already read from disk because the constructor of this class loads the named PNG image from the given png filename.



This is the second example:

```
example12.cpp: """
#include <stdio.h>
#include <stdlib.h>
#include <FL/Fl.H>
#include <FL/Fl_Shared_Image.H>
#include <FL/Fl_Double_Window.H>
#include <FL/Fl_Scroll.H>
#include <FL/Fl_JPEG_Image.H>
#include <FL/Fl_Box.H>
constexpr const int WIDTH = 500;
constexpr const int HEIGHT = 400;
constexpr const char * JPGFILE = "jolly.jpg"; // any .jpg (500 x 400)px will do.
int main() {
  fl_register_images();
  Fl_Double_Window win(WIDTH,HEIGHT,"Example image viewer with scrollbars");
  win.color(FL_WHITE);
  win.begin();
    Fl_Scroll scr(0,0,WIDTH,HEIGHT);
    Fl_JPEG_Image jpg(JPGFILE);
    if ( jpg.h() == 0 ) {
      // error check failed.
      perror(JPGFILE);
      exit(1);
    }
    Fl_Box box(0,0,jpg.w(),jpg.h());
  win.end();
```

```
  box.image(jpg);
  win.resizable(win);
  win.show();
  return(Fl::run());
}
"""
```

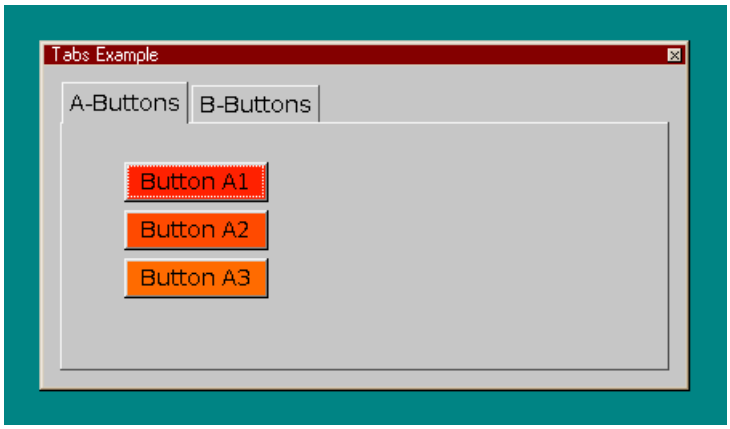
In this example the `Fl_Scroll` widget is used. This is a container widget and will let scrollbars appear if its child widget, here the JPEG image, is larger than the size of this scroll widget. In this program the size of the scroll widget is set to the size of the main window.

The main window is defined using the `Fl_Double_Window` subclass. This subclass will be using a double-buffered window. If possible this window widget will use the X double buffering extension (Xdbe). If not, it will draw the window data into an off-screen pixmap, and then copy it to the on-screen window.

You can take any PNG or JPEG image. I made Google image search for "Jolly Roger" and converted the result into a PNG and a JPEG file for this example.

14. Grouping widgets in Tabs

This is another example from Greg Ercolano showing how TABs work in FLTK. There are two TABs which group two different sets of buttons.




```
example13.cpp: """
#include <FL/Fl.H>
#include <FL/Fl_Window.H>
#include <FL/Fl_Tabs.H>
#include <FL/Fl_Group.H>
#include <FL/Fl_Button.H>
constexpr const int W = 400;
constexpr const int H = 200;
constexpr const int M = 10;
int main(int argc, char *argv[]) {
  Fl_Window *win = new Fl_Window(W,H,"Tabs Example"); {
  // implicit: win->begin();
    Fl_Tabs *tabs = new Fl_Tabs(M,M,W-2*M,H-2*M); {
```

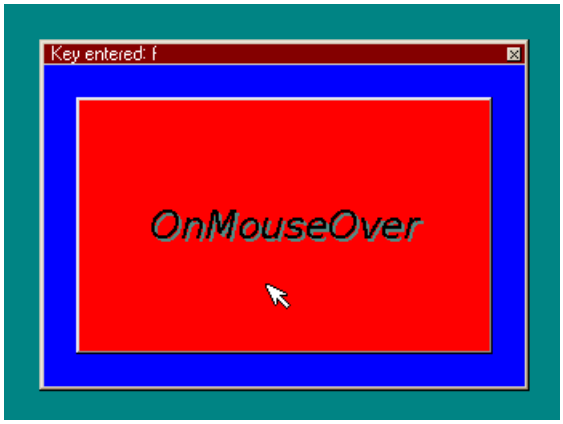
```
    // implicit: tabs->begin();
      // A tab
      Fl_Group *aaa = new Fl_Group(M,3*M+5,W-2*M,H-4*M-5,"A-Buttons"); {
      // implicit: aaa->begin();
        Fl_Button *b1 = new Fl_Button(50, 60,90,25,"Button A1");
        b1->color(88+1);
        Fl_Button *b2 = new Fl_Button(50, 90,90,25,"Button A2");
        b2->color(88+2);
        Fl_Button *b3 = new Fl_Button(50,120,90,25,"Button A3");
        b3->color(88+3);
      } aaa->end();
      // B tab
      Fl_Group *bbb = new Fl_Group(M,3*M+5,W-2*M,H-4*M-5,"B-Buttons"); {
      // implicit: bbb->begin();
        Fl_Button *b1 = new Fl_Button( 50,60,90,25,"Button B1");
        b1->color(88+1);
        Fl_Button *b2 = new Fl_Button(150,60,90,25,"Button B2");
        b2->color(88+3);
        Fl_Button *b3 = new Fl_Button(250,60,90,25,"Button B3");
        b3->color(88+5);
      } bbb->end();
    } tabs->end();
  } win->end();
  win->show();
  return(Fl::run());
}
"""
```

In the examples we have used so far we declared a window object and our buttons and other widgets were in the group of this window object. Here the window has one child which is the TABs widget. This widget has two `Fl_Group` object groups as its childen, the group "aaa" and "bbb". Each of these groups has three buttons as its children again.

The TABs widget will just display the group belonging to the currently selected TAB.

15. Handling mouse events part1

In this program it is monitored whether the mouse is over the box in the middle of the window. If yes, the box will turn red. Keyboard events are also retrieved and displayed in the title bar of the window.



```
example14.cpp: """
#include <FL/Fl_Window.H>
#include <FL/Fl_Double_Window.H>
#include <FL/Fl.H>
#include <FL/Fl_Box.H>
constexpr const int WIDTH = 300;
constexpr const int HEIGHT = 200;
constexpr const int MARG = 20;
Fl_Window* win;
class EventWindow: public Fl_Box {
  int handle_key(int e,int key);
public:
  EventWindow(int t, int l, int width, int height);
  int handle(int e);
};
EventWindow::EventWindow(int t, int l, int width, int height) :
  Fl_Box(FL_UP_BOX, t, l, width, height, "") {
  labelfont(FL_ITALIC);
  labelsize(24);
  labeltype(FL_SHADOW_LABEL);
  label("OnMouseOver");
  Fl::focus(this);
}
int EventWindow::handle(int e) {
  switch(e) {
  case FL_ENTER:
    color (FL_RED);
    labelcolor(FL_BLACK);
    damage(1);
    return 1;
  case FL_LEAVE:
    color(FL_GRAY);
    labelcolor(FL_BLACK);
    damage(1);
    return 1;
  case FL_KEYDOWN:
    return handle_key(e,Fl::event_key());
  default:
    return Fl_Box::handle(e);
  };
}
int EventWindow::handle_key(int e,int key) {
  char labeltext[20];
  sprintf(labeltext,"Key entered: %c  ",key);
  win->label(labeltext);
  return 1;
}
int main() {
  win = new Fl_Window(WIDTH, HEIGHT);
  win->color(FL_BLUE);
  // implicit win->begin();
    EventWindow ewin(MARG, MARG, WIDTH -2*MARG, HEIGHT-2*MARG);
    ewin.show();
  win->end();
  win->show();
  return Fl::run();
}
"""
```

In the `main()` function a new window object is created and this time there is not a button or a box object as before but a new window object called "`ewin`" is added as the child. This window is created using the class `EventWindow` which is defined in this example as a subclass of the `Fl_Window` class. The constructor of this class will write the text "OnMouseOver" into this window.

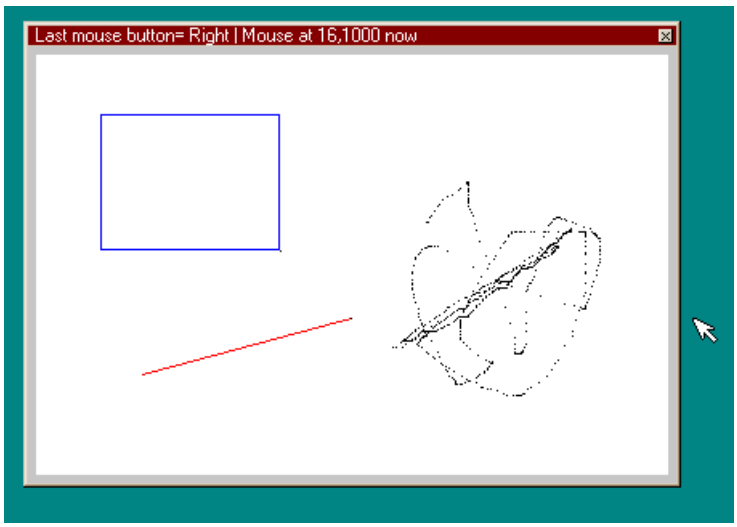
This class also has a method that overwrites the `Fl_Widget::handle()` virtual method. This way all the events are sent to this handle() method by FLTK. The method checks for three different events and returns a one for those to indicate to FLTK that these events are handled by this function: `FL_ENTER`, `FL_LEAVE` and `FL_KEYDOWN`. A `FL_ENTER` event occurs when the mouse is over the window created by the "`ewin`" object while a `FL_LEAVE` event occurs when the mouse has been moved outside this window. These events are used here to change the color of this window.

When a `FL_KEYDOWN` event occured, the function `handle_key()` will be called which displays the key value in the title bar of the main window.

Retrieving keyboard events is usually not necessary with FLTK since you will be using input boxes etc. for keyboard input which do not require you to keep track of every keystroke.

16. Handling mouse events part2

The example is somewhat similar to a paint program. When you click and drag the mouse, a line will be painted following the mouse movement. When you click once a line will be drawn to the point of the next click with the LEFT mouse button. If you click with the RIGHT mouse button a frame will be drawn instead.



This example is based on one posted by Ian MacArthur. Drawing with FLTK is usually done inside the virtual draw() method. You then implement your own version of this method for drawing that overwrites FLTK's `draw()` virtual method. Whenever `draw()` is called it will draw the screen again as defined in this method.

For incremental drawing you better draw into an offscreen buffer that is copied to the

screen whenever the `draw()` method is called. Since FLTK does not store the screen contents these will be erased when the window is iconized and restored. The contents in the offscreen buffer will be preserved though. This example does incremental drawing and shows the use of such an offscreen buffer.

```
example15.cpp: """
#include <Fl/Fl.H>
#include <FL/Fl_Double_Window.H>
#include <Fl/Fl_Box.H>
#include <Fl/fl_draw.H>
#include <stdlib.h>
constexpr const int window_size = 400;
constexpr const int margin = 10;
static Fl_Double_Window *main_window = 0;  // the main app window
static Fl_Offscreen offscreen_buffer = 0;  // the offscreen surface
/*****************************************************************************/
/* This class provides a view to copy the offscreen surface to */
class canvas : public Fl_Box {
  void draw();
  int handle(int event);
public:
  canvas(int x, int y, int w, int h);
};
/*****************************************************************************/
/* Constructor */
canvas::canvas(int x, int y, int w, int h) : Fl_Box(x,y,w,h){}
/*****************************************************************************/
/* draw method */
void canvas::draw() {
  if(offscreen_buffer) { // offscreen exists
    // blit the required view from the offscreen onto the box
    fl_copy_offscreen(x(), y(), w(), h(), offscreen_buffer, 0,0);
  } else {
    // create the offscreen
    main_window->make_current(); //ensures suitable graphic context
    offscreen_buffer = fl_create_offscreen( w(), h() );
    if(!offscreen_buffer) {
      fprintf(stderr,"Failed buffer creation");
      exit(1);
    }
    fl_begin_offscreen(offscreen_buffer); /* Open the offscreen context */
    fl_color(FL_WHITE);
    fl_rectf(0, 0,  w(), h() );
    fl_end_offscreen(); /* close the offscreen context */
    /* init screen with offscreen buffer */
    fl_copy_offscreen(x(), y(), w(), h(), offscreen_buffer, 0,0);
  }
}
/*****************************************************************************/
/* handle */
int canvas::handle(int event) {
  static char labeltext[100];
  int button,x,y;
  int retvalue = 0;
  static int x_old,y_old;
  static int push1st=0;
  if (!offscreen_buffer) return 1;
  retvalue = Fl_Box::handle(event);
```

```
switch (event) {
case FL_PUSH:
case FL_DRAG:
  button = Fl::event_button();
  x = Fl::event_x();
  y = Fl::event_y();
  break;
default:
  button = 0;
  break;
}
switch ( button ) {
case 1: // Left button
  sprintf(labeltext,"Last mouse button= Left | Mouse at %d,%d now",x,y);
  window()->label(labeltext);
  retvalue = 1;
  break;
case 3: // Right button
  sprintf(labeltext,"Last mouse button= Right | Mouse at %d,%d now",x,y);
  window()->label(labeltext);
  retvalue = 1;
  break;
}
switch(event) {
case FL_PUSH:
  if (push1st == 0) {
    x_old = x;
    y_old = y;
    push1st = 1;
    break;
  } else {
    push1st = 0;
    /* Open the offscreen context for drawing */
    fl_begin_offscreen(offscreen_buffer);
    if (button==1) { //left mouse button
      fl_color(FL_RED);
      fl_line(x_old,y_old,x,y);
    } else {
      //right mouse button
      fl_draw_box(
        FL_BORDER_FRAME,x_old,y_old,(x-x_old), (y-y_old),FL_BLUE
      );
    }
    fl_end_offscreen(); /* close the offscreen context */
    redraw();
  }
case FL_DRAG: {
    push1st=0; //clear if dragging
    /* Open the offscreen context for drawing */
    fl_begin_offscreen(offscreen_buffer);
    fl_color(FL_BLACK);
    fl_point(x,y);
    fl_end_offscreen(); // close the offscreen context
    redraw();
  }
  break;
default:
  redraw();
  break;
```

```
  }
  return retvalue;
}
/*****************************************************************************/
int main(int argc, char **argv) {
  main_window = new Fl_Double_Window(
    window_size, window_size,
    "Drawing with mouse example"
  );
  main_window->begin();
    // a view of the offscreen, inside the main window
    static canvas *os_box = new canvas(
      margin,margin,(window_size-2*margin),(window_size-2*margin)
    );
  main_window->end();
  main_window->resizable(os_box);
  main_window->show();
  return Fl::run();
}
"""
```

A new window class called canvas is defined in this example as a subclass of `Fl_Box`. This defines the area that will be used for drawing. A new object of this canvas class is created as a child widget of the `main_window` in the `main()` function.

The canvas class also defines a `handle()` method which overwrites the `Fl_Widget::handle()` virtual method. This way all the events are sent to the canvas `handle()` method by FLTK. This method checks for two different mouse events and returns a one for those to indicate to FLTK that they are handled by this function: `FL_PUSH` and `FL_DRAG`. If one of these events is received, this method will write into the title of the window which mouse button has been clicked last and what the current position of the mouse is.

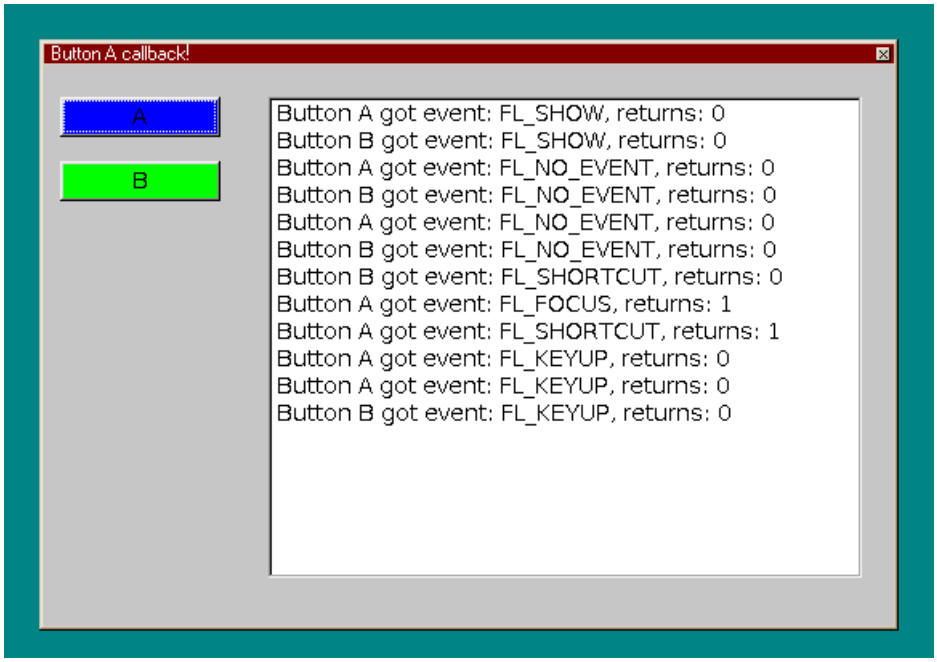
Then, if the event is a `FL_DRAG` it will draw a pixel at the current mouse position. If the event is a `FL_PUSH`, just the mouse position will be stored. On a second `FL_PUSH` event it depends whether the left or the right mouse button has been clicked. If the left mouse button was clicked, a line will be drawn from the position of the first click to the current mouse position. If the click was done with the right mouse button, a box will be drawn instead of the line. The drawing to the offscreen buffer has to be done after a call to `fl_begin_offscreen()` and end with a call to `fl_end_offscreen()`.

The draw method just copies the offscreen buffer to the screen. When it is called for the first time it will use `fl_create_offscreen()` to allocate the offscreen buffer and then use `fl_rectf()` to clear the buffer to white. When the buffer is drawn to the screen it will turn the drawing area to white. Just to ensure we have a suitable graphic context to base the offscreen on the `main_window` is made current.

17. Displaying the events generated by FLTK

This example is based on the one given by Robert Arkiletian in his tutorial. There are two buttons defined in the window which will cause events when the mouse moves over them or clicks them, also keyboard events etc. will be displayed in a scrolling FLTK browser

window. The callback functions defined for the buttons are called when the buttons are clicked or the letters 'a' or 'b' are entered.



The size of the example could be reduced by using the new `fl_eventnames` array in FLTK version 1.3.0.

```
example16.cpp: """
#include <FL/Fl.H>
#include <FL/Fl_Window.H>
#include <FL/Fl_Browser.H>
#include <FL/Fl_Button.H>
#include <FL/names.h>
constexpr const int WIDTH = 530;
constexpr const int HEIGHT = 350;
Fl_Window *win;
Fl_Browser *b;
class MyButton : public Fl_Button {
  static int count;
public:
  char linetext[100];
  MyButton(int x,int y,int w,int h,const char*l=0) :Fl_Button(x,y,w,h,l) {}
  int handle(int e) {
    int ret = Fl_Button::handle(e);
    sprintf(
      linetext, "Button %s got event: %s, returns: %d ",
      label(), fl_eventnames[e], ret
    );
    b->add(linetext);
    b->bottomline(32000); //always display the last line
    return(ret);
  }
};
int MyButton::count=0;
void but_a_cb(Fl_Widget* w, void* v) {
  win->label("Button A callback!");
}
```

```
void but_b_cb(Fl_Widget* w, void* v) {
  win->label("Button B callback!");
}
int main() {
  win= new Fl_Window(WIDTH,HEIGHT,"FLTK events example");
  win->begin();
  b = new Fl_Browser(140,20,win->w()-160, win->h()-50);
  b->type(FL_MULTI_BROWSER);
  MyButton but_a(10,20,100,25,"A");
  but_a.color(FL_BLUE);
  but_a.shortcut('a');
  but_a.callback(but_a_cb);
  MyButton but_b(10,60,100,25,"B");
  but_b.color(FL_GREEN);
  but_b.shortcut('b');
  but_b.callback(but_b_cb);
  win->end();
  win->show();
  return(Fl::run());
}
"""
```

In the `main()` function a new window object is created and a multi browser widget plus two buttons are added as children. The browser widget will list the events generated by FLTK on the screen. For each button a callback function is defined which will display a message in the window's title bar when clicked or when selected by a shortcut key. Shortcut keys are just the letters 'a' and 'b' here, not an ALT- or CTRL- combination.

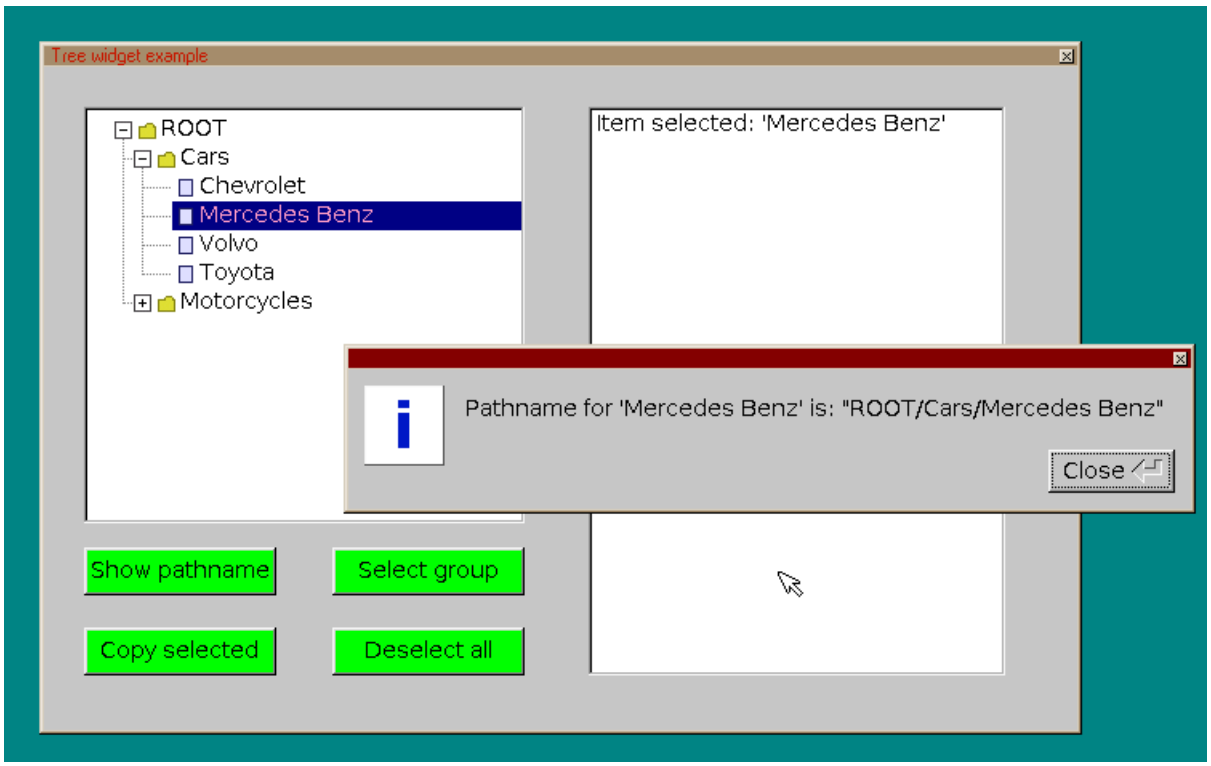
To create the button objects a new class has been defined called "MyButton" which is a subclass of the `Fl_Button` class. All that this class does is to overwrite the `Fl_Widget::handle()` virtual method. This way all the events are sent to this `handle()` method by FLTK and can then be displayed in the browser window. As you can see in the screenshot, some events are sent to both buttons. If the `handle()` function would return a one, the event would be considered handled by FLTK and not be send to other widgets to give these a chance to process the event.

18. The tree widget

There is a new widget in FLTK 1.3.0 developed by Greg Ercolano. It is the tree widget which lets you design file explorer style windows.

The following example allows to select one or more items in the tree by clicking on them. Then by clicking on the "Copy selected" button the selected items will be added to the browser window on the right.

You can also select a single group and by using the "Select group" button select all the items which belong to this group. The items will also be selected when the group display is closed.



By clicking on the "Show pathname" button a message window appears specifying the path of the selected item. Finally the "Deselect all" button removes all selections.

```
example17.cpp: """
#include <FL/Fl.H>
#include <FL/Fl_Double_Window.H>
#include <FL/Fl_Tree.H>
#include <FL/Fl_Button.H>
#include <FL/Fl_Browser.H>
#include <FL/fl_message.H>
constexpr const int W = 645;
constexpr const int H = 414;
Fl_Double_Window* win = new Fl_Double_Window(W, H,"Tree widget example");
Fl_Tree* tree = new Fl_Tree(25, 25, 275, 260);
Fl_Browser *b;
/** Assign user icons to the items */
void AssignUserIcons() {
  static const char *L_folder_xpm[] = {
    "11 11 3 1",
    ".  c None",
    "x  c #d8d833",
    "@  c #808011",
    "...........",
    ".....@@@@..",
    "....@xxxx@.",
    "@@@@@xxxx@@",
    "@xxxxxxxxx@",
    "@xxxxxxxxx@",
    "@xxxxxxxxx@",
    "@xxxxxxxxx@",
    "@xxxxxxxxx@",
    "@xxxxxxxxx@",
    "@@@@@@@@@@"
  };
```

```
  static Fl_Pixmap L_folderpixmap(L_folder_xpm);
  static const char *L_document_xpm[] = {
    "11 11 3 1",
    ".  c None",
    "x  c #d8d8f8",
    "@  c #202060",
    ".@@@@@@@@@.",
    ".@xxxxxxx@.",
    ".@xxxxxxx@.",
    ".@xxxxxxx@.",
    ".@xxxxxxx@.",
    ".@xxxxxxx@.",
    ".@xxxxxxx@.",
    ".@xxxxxxx@.",
    ".@xxxxxxx@.",
    ".@@@@@@@@@."
  };
  static Fl_Pixmap L_documentpixmap(L_document_xpm);
  // Assign user icons to tree items
  for ( Fl_Tree_Item *item = tree->first(); item; item=item->next()) {
    // Assign custom icons
    item->usericon(item->has_children() ? &L_folderpixmap : &L_documentpixmap);
    //item->usericon(0); // Don't assign custom icons
  }
  tree->redraw();
}
void tree_cb(Fl_Tree* w, void*) {
  Fl_Tree *tree = (Fl_Tree*)w;
  // Find item that was clicked
  Fl_Tree_Item *item = (Fl_Tree_Item*)tree->item_clicked();
  if ( item->is_selected() ) {
    //item is already selected
  } else {
    //click on additional item
    tree->select(item); // select this one too
  }
  AssignUserIcons(); //includes tree->redraw();
}
//Now the button callbacks for four buttons
void button_pathname_cb(Fl_Widget*) {
  char pathname[256];
  Fl_Tree_Item *item = tree->first_selected_item();
  if ( !item ) {
    fl_message("No item was selected");
  } else {
    tree->item_pathname(pathname, sizeof(pathname), item);
    fl_message(
      "Pathname for '%s' is: \"%s\"",
      (item->label() ? item->label() : "???"),
      pathname
    );
  }
}
void button_copy_cb(Fl_Widget*) {
  char linetext[100];
  for ( Fl_Tree_Item *item = tree->first(); item; item = tree->next(item)) {
    if ( item->is_selected()) {
      sprintf(linetext,"Item selected: '%s'\n", item->label());
```

```
      b->add(linetext);
      b->bottomline(32000); //always display the last line
    }
  }
}
void button_select_group_cb(Fl_Widget*) {
  Fl_Tree_Item *item = (Fl_Tree_Item*)tree->first_selected_item();
  if (item && item->has_children()) {
    tree->select_all(item);
    tree->deselect(item,0);
      //do not select the folder itself
      //2nd parameter=0 - no callback triggered
    AssignUserIcons(); //includes tree->redraw();
  } else {
    fl_message("Please select a folder first!");
  }
}
void button_deselect_cb(Fl_Widget*) {
  tree->deselect_all(0,0); //2nd parameter=0 - no callback triggered
  AssignUserIcons(); //includes tree->redraw();
}
int main(int argc, char **argv) {
  // Create tree and add items
  // implicit: tree->begin();
    tree->add("Cars/Chevrolet");
    tree->add("Cars/Mercedes Benz");
    tree->add("Cars/Volvo");
    tree->add("Cars/Toyota");
    tree->add("Motorcycles/Harley Davidson");
    tree->add("Motorcycles/Honda");
    tree->add("Motorcycles/Suzuki");
    tree->close("/Motorcycles");
    tree->selectmode(FL_TREE_SELECT_MULTI); // Multiple Items
    tree->callback((Fl_Callback*)tree_cb);
  tree->end();
  //Create buttons and define callbacks
  Fl_Button *button_pathname = new Fl_Button(25,300,120,30,"Show pathname");
  button_pathname->color(FL_GREEN);
  button_pathname->callback(button_pathname_cb);
  Fl_Button *button_select_group = new Fl_Button(180,300,120,30,"Select group");
  button_select_group->color(FL_GREEN);
  button_select_group->callback(button_select_group_cb);
  Fl_Button *button_copy = new Fl_Button(25,350,120,30,"Copy selected");
  button_copy->color(FL_GREEN);
  button_copy->callback(button_copy_cb);
  Fl_Button *button_deselect = new Fl_Button(180,350,120,30,"Deselect all");
  button_deselect->color(FL_GREEN);
  button_deselect->callback(button_deselect_cb);
  //Create browser window
  b = new Fl_Browser(340,25,260,355);
  b->type(FL_MULTI_BROWSER);
  win->end();
  // Display the window
  win->show();
  AssignUserIcons(); //includes tree->redraw();
  // Run and return
  return Fl::run();
}
"""
```

The main window object, the tree widget object and the browser object are defined as globals here to simplify access to them from the callback functions.

In the main function the tree widget is set up with the tree_cb as its callback. This callback just allows to select several items simultaneously which is possible since the browser type is set to `FL_MULTI_BROWSER`.

Then four buttons are defined with separate callbacks. The "Show pathname" button callback retrieves the first selected item in the tree and then uses the `item_pathname()` method to get its pathname which it then displays in a message window.

The "Copy selected" button callback walks in a loop through all items in the tree and if one of them is selected it will copy that to the browser window.

The "Select group" button callback looks if the first selected item in the tree is a folder, i.e. it has children, and if yes will select all items belonging to the folder. The folder item itself will be deselected again in this example.

The "Deselect all" button callback uses the `deselect_all()` method to deselect all items in the tree.

The AssignUserIcons function at the beginning of the code is frequently called in this example. This is taken from the "Tree" example in the FLTK package. It defines two icons using the XPM format and these will be added to the connector line depending whether it is a folder or a file. Then it calls `tree_redraw()` to refresh the tree display. These icons just make the tree look better, they are not required.

19. References

We read the following resources and got most information from:
(All references last checked on 10 sept 2026, unreachable references kept in recognition of original authors.)

a) Robert Artiletian's tutorial:

~~http://www3.telus.net/public/robark/~~ (dead link)

b) Greg Ercolano's FLTK Cheat page:

http://seriss.com/people/erco/fltk/

c) Tutorials on the National Taiwan University site:

~~http://graphics.csie.ntust.edu.tw/courses/index.php/Main/Tutorial~~ (404 not found)

d) The FLTK manual:

http://www.fltk.org/doc-1.3/index.html
http://www.fltk.org/doc-1.4/index.html
http://www.fltk.org/doc-1.5/index.html

e) Other FLTK related sources:

https://www.fltk.org/
https://github.com/fltk/fltk
https://www.fltk.org/newsgroups.php
https://github.com/fltk/fltk/discussions

f) Other references for tools and packages (use: depending on context).

https://www.dev-cpp.com/Dev-Cpp-5/
http://code.google.com/p/fltkwinbin/downloads/list
http://gintasdx.althirius-studios.com/2011/08/tutorial-codeblocks-with-fltk.html
http://code.google.com/p/nanox-microwindows-nxlib-fltk-for-dos/downloads/list

20 Changes.

2026/09/18, AS.
-Added 'small prints' with version tracking and other necessary information.
-Expanded, checked and marked 'References' and all other hyperlinks in the text.
-Added note to use `<Ctrl>+`, `<Ctrl>-` and environment variable `FLTK_SCALING_FACTOR` for scaling the gui appearance.
-All examples numbered (in sequence of appearance) and as a 'multiline sting' for automatic sources extraction.
-All sources validated with Boost test and heaptrack (on debian trixie). The function `main(…)` returns 0 in all examples.

Code specific changes introduced:
Use of '`constexpr cons`' over magic numbers. Makes it easier to experiment with the examples.

FLTK uses `` `f(… void * data, …)' `` for typeless argument passing of data and then casting, e.g.: `` `(int)data' ``. This is 'by (historical) design' in FLTK. While an excepted practice in the past the casting is now flagged as a (potential) problem by most compilers. Casting replaced, for ints with `` `static_cast<int>(reinterpret_cast<intptr_t>((data)))' ``

If you have a tool to convert to plain text then the following script can extract all examples from a text stream. Some (most?) converters replace multiple whitespace characters with one. A good tool for conversion is pandoc. A good tool for C/C++ reformatting is clang-format.

```
extract.awk: """
BEGIN { in_code = 0 }
/^example[0-9]+\.cpp: """$/ { in_code = 1; $0 = "// " $0 }
/^"""$/ && (incode != 0) { in_code = 0; print "// " $0 "\n" }
(incode != 0) { print }
"""
```

2012/03/06, GP.
intitial version.